THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The decimal expansion of $\frac{129}{312500}$ is: 1

(a) terminating **(b)** non-terminating repeating
(c) non-terminating non-repeating **(d)** an integer

2. If $\sqrt{5}$ is irrational, then $\sqrt{5} \times \sqrt{5}$ is: 1

(a) rational **(b)** not defined
(c) non-terminating non-repeating **(d)** irrational

3. The HCF of 56, 84 and 140 is: 1

(a) 28 **(b)** 29 **(c)** 27 **(d)** 30

4. The LCM of 9, 12 and 18 is: 1

(a) 35 **(b)** 37 **(c)** 36 **(d)** 194

5. The largest number that divides 110 and 170 leaving remainders 5 and 2 respectively is: 1

(a) 20 **(b)** 23 **(c)** 21 **(d)** 22

6. The least number which when divided by 9, 12 and 15 leaves remainder 4 each time is: 1

(a) 180 **(b)** 184 **(c)** 185 **(d)** 176

7. The decimal expansion of $\frac{35}{50}$ is: 1

(a) non-terminating non-repeating **(b)** non-terminating repeating
(c) terminating **(d)** an integer

8. If $\sqrt{7}$ is irrational, then $\sqrt{7} \times \sqrt{7}$ is: 1

(a) non-terminating non-repeating **(b)** irrational
(c) rational **(d)** not defined

9. The HCF of 72, 108 and 180 is: 1

(a) 35 **(b)** 36 **(c)** 37 **(d)** 38

10. The LCM of 12, 15 and 20 is: 1

(a) 59 **(b)** 61 **(c)** 360 **(d)** 60

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest number which leaves remainder 6 when divided by 15, 20 and 25. 2

12. Without actual division, find how many decimal places $\frac{7}{32}$ has. 2

13. Find the largest number which divides 615 and 963 leaving remainders 3 and 6 respectively. 2

14. Find the smallest number which leaves remainder 3 when divided by 14, 21 and 35. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The HCF of two numbers is 8 and their LCM is 240. If one number is 48, find the other. 3

16. Three bells toll at intervals of 8, 20 and 30 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

17. Show that $(6 + 1\sqrt{5})(6 - 1\sqrt{5})$ is rational but $6 + 1\sqrt{5}$ is irrational. 3

OR

The HCF of two numbers is 15 and their LCM is 450. If one number is 90, find the other. 3

18. Three bells toll at intervals of 9, 12 and 21 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

OR

Show that $(8 + 2\sqrt{7})(8 - 2\sqrt{7})$ is rational but $8 + 2\sqrt{7}$ is irrational. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two tankers contain 420 litres and 546 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

A rectangular courtyard is 72 m by 48 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

20. Two tankers contain 360 litres and 504 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

A rectangular courtyard is 96 m by 60 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

END OF QUESTION PAPER — Total: 40 Marks

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